

the engine beam of $V = 0.34 \mu\text{m}^3$, has a power density $p_{\text{beam}} = \dot{W}_e/V = 37 \text{ GW/m}^3$ which is almost a factor 1000 higher than that of car engines which have typical power densities of around 50 MW/m^3 . The power density p_{beam} of the piezoresistive heat engine is also high when compared to other microengines^{1,20,21}, which have reported power densities up to 3 GW/m^3 . However, in some of these cases the power density is defined using the volume of the total structure instead of the engine displacement volume. Due to the relatively large size of the proof mass compared to the engine beam size, this definition of power density is less favourable for the piezoresistive heat engine, yielding $p_{\text{tot}} = 11 \text{ MW/m}^3$.

Using the equations from Fig. 5a, the efficiency of the presented engine is calculated to be $\eta_{he} = (\dot{W}_{he} - \dot{Q}_{fd})/\dot{Q}_r = 6.4 \times 10^{-7}$. Besides being limited by the Carnot theorem, the efficiency of the piezoresistive heat engine is also limited by the fracture limit of silicon to around $\eta_{pr, Si, \text{max}} = 3.4 \times 10^{-4}$, as analysed in the Supplementary Discussion B. It is therefore expected that the efficiency and power density of the piezoresistive heat engine can still be improved by a factor ~ 500 by increasing the displacement amplitude and thermal design of the engine. Even after optimization, this efficiency is low compared to that of micro- and nano-engines driven by electrostatic forces^{22–25}. Nevertheless, the comparison in Supplementary Discussion B shows that piezoresistive heat engines have the potential to outperform electrostatic engines in terms of power density. This high power density and the absence of transistors can facilitate the development of mechanical actuator²⁶ and sensor^{10,27} arrays with very high element densities.

Similar to other recently investigated cooling methods^{11–18}, the coefficient of performance of the refrigerator $\eta_{\text{cool}} = -\dot{W}_{he}/\dot{Q}_r$ is very small when T_{eff} is at or below room temperature. The refrigerator will therefore not be useful for lowering the temperature of systems with many degrees-of-freedom. However, because the heat capacity $\frac{\partial U_i}{\partial T_{\text{eff}}}$ of a single-degree-of-freedom resonance mode is only $\frac{1}{2}k_B$ and because the thermal insulation of the mode is high as a result of its high Q_{int} (see Fig. 5a), the piezoresistive refrigerator can still reduce T_{eff} by more than 200 K. The Brownian fluctuations associated with the energy $\frac{1}{2}k_B T_{\text{eff}}$ often dominate the noise in mechanical sensors and mirrors and their reduction can thus increase the signal-to-noise ratio and might ultimately enable quantum-limited measurement and control of mechanical resonators. The presented piezoresistive refrigerator provides an alternative cooling method for reducing these Brownian fluctuations, which has the advantage that it does not require laser or RF sources and only requires a silicon chip with a relatively low DC power.

Methods

Structure The piezoresistive heat engine in Fig. 1a is made out of a $1.5 \mu\text{m}$ thick silicon layer on a silicon-on-insulator wafer, with a phosphor doping concentration $N_d = 4.5 \times 10^{18} \text{ cm}^{-3}$ giving a specific resistivity of $\rho_{dc} = 10^{-4} \Omega\text{m}$. This doping concentration results in a low temperature coefficient of resistivity²⁸ of less than 0.1% per K (see Supplementary Fig. C). The thin crystalline silicon layer is structured in a single mask step by a deep reactive ion etch and the buried SiO_2 layer below the mass and beams is removed in a hydrogen fluoride vapour etch. A DC current I_{dc} is driven through the beams via terminals T1 and T2 as shown in Fig. 1b and c. As a result of the geometry of the structure, the current density, heating power density and mechanical stress are concentrated in the narrow engine beam, but the resonance frequency of the resonator is mainly determined by the spring beam and resonator mass. The in-plane mechanical bending resonance mode that determines the fundamental resonance frequency $f_0 = 1.26 \text{ MHz}$ has a spring constant $k = 256 \text{ N/m}$ as determined from finite element method (FEM) simulations. Measurements are performed at 315 K in a vacuum chamber at a pressure below 10^{-2} mbar .

External excitation The in-plane fundamental mechanical resonance of the structure at 1.26 MHz is characterized by actuating it via terminal T3 with an AC electrostatic force $F_{ac, \text{act}} = \epsilon_0 A V_{dc, \text{act}} v_{ac, \text{act}} / g^2$ generated by a voltage $V_{dc, \text{act}} + v_{ac, \text{act}}$ on terminal T3, with $V_{dc, \text{act}} = -1 \text{ V}$. The actuation gap $g = 200 \text{ nm}$ has an actuation area $A = 60 \times 1.5 \mu\text{m}^2$. A DC current $I_{dc} = 0.1 \text{ mA}$ is applied as in Fig. 1b. The displacement of the centre of mass x is proportional to the measured piezoresistive AC current i_{pr} , which is shown in Fig. 2a. The reduction of the intrinsic Q -factor Q_{int} upon increasing the pressure in the vacuum chamber confirms the mechanical nature of the resonance. The solid fits in Fig. 2a allow us to determine the piezoresistive factor K_{pr} , which is defined by $r_{ac}/R_{dc} = K_{pr}x$, where r_{ac} is the amplitude of the piezoresistance variation and R_{dc} is the total DC resistance. From the maximum piezoresistive current $i_{pr, \text{peak}}$ the piezoresistive factor is determined to be $K_{pr} = -\frac{i_{pr, \text{peak}}}{I_{dc}} \frac{k}{F_{ac, \text{act}} Q_{\text{int}}} = -6.6 \times 10^5 \text{ m}^{-1}$.

Spectrum analysis The spectra in Fig. 3 and 4 are measured using the electrical circuit in Fig. 1c. Because the displacement of the centre-of-mass $x_\omega \equiv x(\omega)$ causes a piezoresistive voltage $v_\omega \equiv v_{ac} = [Z_{\text{tot}}^{-1} + R_{dc}^{-1}]^{-1} I_{dc} K_{pr} x$ (Fig. 1c), the electrical spectrum $\langle v_\omega^2 \rangle$ measured by the spectrum analyser is proportional to equation (1). The voltage spectral density in Fig. 3a and b is fit by dashed curves which are given by equation (1) with $\langle v_\omega^2 \rangle = I_{dc}^2 K_{pr}^2 \langle x_\omega^2 \rangle / [Z_{\text{tot}}^{-1} + R_{dc}^{-1}]^2$ and which include a fit parameter to account for the white Johnson noise background. The total AC impedance parallel to the resonator $Z_{\text{tot}} = (i\omega C_0)^{-1} + (i\omega C_1 + Z_{\text{load}}^{-1})^{-1}$ can be controlled by adjusting Z_{load} . The DC resis-